

Detecting departure and destination aerodromes from ADS-B

Version 6 — the consolidated study, and what to run in production

EUROCONTROL Performance Review Unit

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Summary

OPDI derives a flight's departure and destination aerodrome from ADS-B state vectors alone. This page consolidates six versions of that work into one account: what the three candidate methods are, how each behaves across its whole parameter space, which settings are best for departures and for arrivals, what the ideas that did *not* work were, and what the pipeline should ship.

The short version is that the two roles want different algorithms, and that the algorithm already in production was never tuned. Its flight-level gate — the altitude below which state vectors are allowed to vote on whether a track is climbing or descending — was set at FL40 and never swept. Moving it is worth thousands of correctly named flights per three-day sample, at a cost of tens of wrong ones.

Reading this page

Every number below is an inline expression over a CSV in `data/`. Nothing is typed into the prose. Where this page disagrees with versions 1–5, it says so and gives the reason; those versions stay published unchanged.

1 How the metrics work

Five quantities are reported for every setting, for each role separately. Four of them are unavoidable and the fifth is a judgement, so it is worth being precise about all five before any of them are used to argue for a parameter.

Write N for the number of ground-truth flights in the sample, a for the number the method *answers* (names an aerodrome, or explicitly says out-of-area), c for the number it answers **correctly**, and $w = a - c$ for the number it answers **wrongly**.

Coverage is a/N : the share of flights the method is willing to commit on. A method that abstains everywhere has coverage zero and cannot be wrong.

Accuracy is c/a : of the flights it committed on, the share it got right. Accuracy says nothing about how often the method declined to answer, so a method can buy accuracy simply by abstaining more.

Correct (c) and **wrong** (w) are the raw counts. They matter because coverage and accuracy generally move in opposite directions — every extra answer a method gives is one it was previously unsure about, so raising coverage tends to lower accuracy — and neither ratio says whether a particular exchange was worth making. The counts do.

Score is $c - kw$. It exists to make the exchange rate explicit: a change to a parameter is worth taking if and only if it wins more than k correct answers for each wrong one it introduces.

The value of k is where judgement enters. This study uses $k = 2$ by default, on the following argument. A **silence** — an abstention — corrupts one movement count, and it is *visible*: it arrives downstream as an explicit null that any consumer can filter on. A **wrong aerodrome** corrupts *two* movement counts, the one it invents and the one it omits, and it is *invisible*: nothing downstream can distinguish it from a correct answer. Two errors for one, plus undetectability, is the case for $k = 2$.

That is an argument, not a measurement, and rankings do move with k . So every headline choice on this page is also scored at $k = 1$ and $k = 3$, and where the ranking changes, the page says so.

1.1 Out-of-area, and why the headline numbers are decomposed

Roughly one flight in six in the sample genuinely begins or ends outside the observed area. Such a flight has no nameable aerodrome, and the correct answer is the literal label OOA. This creates a way to score well for a bad reason: a method that labels liberally out-of-area collects those flights for free.

So each side is also reported three further ways — out-of-area recall and precision, which show whether the free wins are being taken honestly, and in-area coverage and accuracy, which restrict to flights whose truth is a real aerodrome and are the numbers to read when comparing naming rules.

Table 1: The reported quantities.

Quantity	Definition	Reads as
Coverage	a/N	willingness to commit
Accuracy	c/a	reliability when committed
Correct	c	flights named right
Wrong	$w = a - c$	flights named wrong
Score	$c - kw$	net value at exchange rate k
OOA recall	OOA hits / OOA in truth	does it recognise what it cannot name?
OOA precision	OOA hits / OOA predicted	is it over-using the OOA label?
In-area coverage	answered / truth is a real aerodrome	commitment with the free wins removed
In-area accuracy	correct / answered, truth a real aerodrome	reliability with the free wins removed

2 The sample

The 2025 period is 5-7 June 2025, three consecutive days, giving **95,116 ground-truth flights** after alignment with the Network Manager reference.

Ground truth is the EUROCONTROL Network Manager flight table, joined to ADS-B tracks on aircraft address, callsign and date. The join key is `AIRCRAFT_ADDRESS`, which is the `icao24` of the ADS-B message.

3 The three methods

3.1 trend — what production runs today

For every state vector below a flight-level cap and within a radius of an aerodrome, **trend** asks whether smoothed barometric altitude is rising or falling. Each vector casts a vote. If climbing votes exceed descending votes by more than a margin, that aerodrome is the departure; if the reverse, the destination. The nearest surviving candidate wins.

Two of its four constants had never been swept before this study: the flight-level cap, fixed at FL40, and the detection radius, fixed at 30 NM.

3.2 nearest — the *traffic* library's rule

The reference implementation in the **traffic** library takes the aerodrome nearest the first and last point of the trajectory, unconditionally. It never abstains, so its coverage is total and its accuracy is whatever the geometry gives. It is included as the external baseline.

3.3 endpoint — nearest, with abstention

endpoint is **nearest** plus two gates: the endpoint must lie within a radius of the aerodrome, and either on the ground or below a height above the field. Failing both, the method abstains rather than guessing. A scheduled-service penalty breaks ties toward aerodromes with commercial traffic.

4 Trend: the full parameter sweep

4.1 The untuned baseline

- **ADEP** at the production setting (FL40, margin 4, 30 NM, no penalty): coverage 68.35%, accuracy 98.73%, 64,188 correct, 826 wrong, score 62,536.
- **ADES** at the production setting (FL40, margin 4, 30 NM, no penalty): coverage 67.38%, accuracy 97.52%, 62,500 correct, 1,589 wrong, score 59,322.

4.2 The best setting found

- **ADEP**: FL120, margin 0, 20 NM, penalty 10 NM — coverage 77.43%, accuracy 98.58%, 72,607 correct, 1,045 wrong, score 70,517 (+7,981 against the untuned baseline).
- **ADES**: FL60, margin 2, 20 NM, penalty 10 NM — coverage 69.50%, accuracy 97.44%, 64,416 correct, 1,689 wrong, score 61,038 (+1,716 against the untuned baseline).

4.3 The full trend grid

trend has three geometric parameters, so the sweep is a cube and cannot be printed as one table. It is sliced instead: for each role, score by flight-level cap against radius at that role's best vote margin, then the margin's own profile, then the penalty. Every cell of the cube is in `data/trend_sweep_2025.csv`.

ADEP — score at $k = 2$, by flight-level cap (rows) and radius in NM (columns), at the winning vote margin of 0

FL cap	5	10	15	20	30	40	60	80
20	56,065	55,966	55,909	55,855	55,725	55,663	55,403	54,601
30	62,657	62,897	62,821	62,746	62,588	62,520	62,198	61,456
40	64,619	65,913	65,830	65,746	65,543	65,451	65,108	64,229
60	65,139	67,927	67,959	67,872	67,621	67,498	67,123	66,283
80	64,942	68,768	69,395	69,347	69,082	68,910	68,487	67,628
100	64,650	68,797	70,080	70,246	69,966	69,723	69,234	68,350
120	64,189	68,383	70,037	70,517	70,339	70,066	69,543	68,634
150	62,892	67,206	68,925	69,464	69,586	69,326	68,727	67,826
200	59,862	63,905	65,286	65,617	65,739	65,530	64,855	63,936

ADES — score at $k = 2$, by flight-level cap (rows) and radius in NM (columns), at the winning vote margin of 2

FL cap	5	10	15	20	30	40	60	80
20	58,185	58,844	58,612	58,471	58,316	58,213	58,064	57,996
30	58,522	60,169	60,076	59,944	59,781	59,666	59,532	59,448
40	58,435	60,290	60,675	60,720	60,573	60,450	60,306	60,209
60	57,654	59,020	59,740	61,038	61,023	60,867	60,652	60,519
80	56,703	58,043	58,717	60,259	60,476	60,292	60,027	59,838
100	54,684	55,990	56,625	58,100	58,595	58,482	58,176	57,916
120	52,961	54,175	54,842	56,035	56,532	56,423	56,086	55,794
150	51,410	52,237	52,637	53,639	53,887	53,788	53,428	53,112
200	48,982	49,205	49,106	49,642	49,346	49,066	48,714	48,383

The vote margin, at each role's winning cap and radius:

Role	Margin	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.43%	98.58%	72,607	1,045	70,517
ADEP	2	77.18%	98.64%	72,409	997	70,415
ADEP	4	76.98%	98.68%	72,251	968	70,315
ADEP	8	76.32%	98.72%	71,665	930	69,805
ADEP	16	74.86%	98.78%	70,333	868	68,597
ADES	0	69.65%	97.36%	64,500	1,752	60,996
ADES	2	69.50%	97.44%	64,416	1,689	61,038
ADES	4	69.11%	97.52%	64,105	1,631	60,843
ADES	8	68.60%	97.59%	63,676	1,574	60,528
ADES	16	67.36%	97.69%	62,591	1,477	59,637

The scheduled-service penalty applied to trend, which production does not do at all, swept at each role’s winning geometry:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.43%	96.37%	70,977	2,675	65,627
ADEP	5	77.43%	98.39%	72,463	1,189	70,085
ADEP	10	77.43%	98.58%	72,607	1,045	70,517
ADEP	15	77.43%	98.44%	72,504	1,148	70,208
ADEP	20	77.43%	98.29%	72,391	1,261	69,869
ADEP	30	77.43%	98.29%	72,391	1,261	69,869
ADES	0	69.50%	95.34%	63,025	3,080	56,865
ADES	5	69.50%	97.14%	64,216	1,889	60,438
ADES	10	69.50%	97.44%	64,416	1,689	61,038
ADES	15	69.50%	97.37%	64,369	1,736	60,897
ADES	20	69.50%	97.28%	64,306	1,799	60,708
ADES	30	69.50%	97.28%	64,306	1,799	60,708

Coverage is constant down each block and only accuracy moves, which is the signature of a pure re-ranking: the penalty changes which aerodrome is named, never whether one is.

5 Endpoint: the full parameter sweep

endpoint has two geometric parameters. The **radius** is how close to an aerodrome the trajectory’s first or last point must be; the **height** is how far above field elevation it may be. Failing either, the method abstains.

The grid below is the whole sweep, scored for each role. The height row labelled *unrestricted* removes the height gate entirely, which collapses **endpoint** into **nearest** within the radius — so the last row of each table is the cost of having no abstention at all.

ADEP — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	28,778	28,709	28,684	28,660	28,655	28,643	28,643	28,643	28,643
500	39,506	39,417	39,358	39,324	39,319	39,307	39,307	39,307	39,307
1,000	52,441	52,334	52,249	52,203	52,199	52,177	52,177	52,177	52,177
2,000	62,124	61,996	61,893	61,828	61,818	61,770	61,766	61,766	61,766
5,000	66,124	68,515	68,366	68,259	68,184	68,070	68,043	68,043	68,043
8,000	66,271	70,079	70,668	70,477	70,368	70,250	70,215	70,213	70,213
10,000	66,246	70,288	71,775	71,591	71,437	71,289	71,242	71,240	71,238
15,000	66,220	70,295	72,591	72,755	72,541	72,261	72,202	72,196	72,194
20,000	66,187	70,158	72,258	72,370	72,144	71,758	71,666	71,648	71,646
25,000	66,142	69,990	71,782	71,470	70,623	69,819	69,656	69,629	69,623
30,000	66,111	69,867	71,159	70,179	68,767	67,474	67,261	67,156	67,149
35,000	66,025	69,221	69,355	66,954	64,171	61,585	61,218	60,979	60,936
40,000	65,990	68,914	68,206	65,014	61,108	57,105	56,349	55,817	55,598
unrestricted	65,993	68,905	68,171	64,951	61,010	56,972	56,200	55,648	55,419

ADES — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	32,816	32,719	32,597	32,449	32,372	32,282	32,266	32,262	32,262
500	48,233	48,127	47,950	47,785	47,706	47,611	47,595	47,587	47,587
1,000	52,528	52,397	52,172	51,975	51,881	51,770	51,746	51,738	51,738
2,000	53,566	54,560	54,321	54,088	53,959	53,833	53,791	53,781	53,781
5,000	53,524	55,632	57,190	56,955	56,735	56,523	56,420	56,410	56,410
8,000	53,607	55,658	57,703	57,797	57,493	57,250	57,119	57,105	57,105
10,000	53,616	55,606	57,663	57,969	57,684	57,414	57,275	57,261	57,261
15,000	53,640	55,513	57,133	57,116	56,759	56,420	56,267	56,251	56,249
20,000	53,697	55,406	56,381	55,330	54,666	54,218	54,039	53,999	53,997
25,000	53,818	55,447	55,946	54,362	53,406	52,565	52,286	52,228	52,222
30,000	53,900	55,400	55,457	53,361	51,914	50,641	50,103	50,010	49,995
35,000	54,022	54,821	53,200	49,348	46,467	43,182	41,661	41,019	40,805
40,000	54,808	55,179	52,368	47,092	42,991	37,190	34,908	33,249	32,673
unrestricted	54,961	55,315	52,464	47,110	42,854	36,840	34,520	32,809	32,217

5.1 Where each role's optimum sits

- **ADEP**: 30 NM, 15,000 ft — coverage 80.96%, accuracy 98.16%, 75,587 correct, 1,416 wrong, score 72,755.
- **ADES**: 30 NM, 10,000 ft — coverage 67.23%, accuracy 96.89%, 61,951 correct, 1,991 wrong, score 57,969.

Both optima are interior to the grid in both parameters, so neither is an artefact of where the sweep stopped.

6 The scheduled-service penalty

Both methods choose among surviving candidates by distance. Near a major aerodrome that often means choosing between the airport itself and a small airfield a few miles away that almost never sees commercial traffic. The penalty adds a fixed distance to every candidate without scheduled service, biasing the tie-break toward the aerodrome more likely to be the real one.

It is a pure re-ranking: it changes *which* aerodrome is named, never *whether* one is named. So coverage is flat across the whole sweep and only accuracy moves — which is exactly what makes it cheap.

Endpoint, at 40 NM and 15,000 ft:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	81.20%	97.31%	75,158	2,077	71,004
ADEP	5	81.20%	97.72%	75,470	1,761	71,948
ADEP	10	81.20%	97.98%	75,667	1,563	72,541
ADEP	15	81.19%	97.90%	75,610	1,619	72,372
ADEP	20	81.20%	97.68%	75,444	1,793	71,858
ADEP	30	81.19%	97.59%	75,365	1,864	71,637
ADES	0	70.00%	93.85%	62,487	4,092	54,303
ADES	5	69.99%	94.31%	62,786	3,787	55,212

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADES	10	69.97%	95.10%	63,287	3,264	56,759
ADES	15	69.92%	94.88%	63,105	3,404	56,297
ADES	20	69.91%	94.80%	63,037	3,456	56,125
ADES	30	69.82%	94.84%	62,985	3,425	56,135

7 How much the exchange rate matters

k is a judgement, so the honest thing is to show what it decides. The table below re-runs the argmax at $k = 1, 2$ and 3 for both methods and both roles.

k	Role	Best trend	trend score	Best endpoint	endpoint score	Winner
1	ADEP	FL120, margin 0, 20 NM, pen 10	71,562	40 NM, 20,000 ft, pen 10	74,233	endpoint
1	ADES	FL80, margin 2, 30 NM, pen 10	62,826	40 NM, 15,000 ft, pen 10	60,023	trend
2	ADEP	FL120, margin 0, 20 NM, pen 10	70,517	30 NM, 15,000 ft, pen 10	72,755	endpoint
2	ADES	FL60, margin 2, 20 NM, pen 10	61,038	30 NM, 10,000 ft, pen 10	57,969	trend
3	ADEP	FL120, margin 0, 20 NM, pen 10	69,472	20 NM, 15,000 ft, pen 10	71,400	endpoint
3	ADES	FL40, margin 2, 15 NM, pen 10	59,482	20 NM, 8,000 ft, pen 10	56,368	trend

The **choice of method is the same at every k tried**: departures want **endpoint** and arrivals want **trend** whether a wrong answer is priced at one silence, two or three. That is the most robust finding on this page, and it does not depend on the judgement that fixes k .

The parameter *values* are less stable than the method choice, and in a direction that makes sense: a higher k prices wrong answers more dearly, so the argmax moves toward tighter gates that answer less often and are right more often when they do. Where a headline value moves with k , the justification table records it.

8 Negative results, re-measured

Ideas that were tried and did not work. All are re-run here against current code rather than quoted from the version that first reported them.

8.1 The approach cone

The box rule treats radius and height as independent, and they are not: an aircraft descending toward an aerodrome trades height for distance at roughly a constant rate. A 3° glideslope is 318 ft/NM and the “three times the flight level” rule of thumb about 333 ft/NM, so the endpoints that genuinely belong to an aerodrome should lie in a *cone*, not a box.

The two shapes disagree in both corners. Close and high is inside the box and outside the cone — an aircraft at 15,000 ft overhead is overflying, not arriving. Far and low is outside the box and inside the cone — an arrival whose reception dies at 8,000 ft and 30 NM is on profile.

Replacing the box with a cone is worse for both roles, at every slope swept.

- **ADEP**: best cone (1000 ft/NM, 40 NM) scores 65,735 against the box’s 72,755, −7,020. It is worse on both counts — fewer correct *and* more wrong, so no exchange rate rescues it.
- **ADES**: best cone (500 ft/NM, 40 NM) scores 57,614 against the box’s 57,969, −355. It is more restrictive, giving up 443 correct to avoid 44 wrong — a trade worth taking only if a wrong answer cost more than 10 silences.

The cone loses for both roles at every slope swept, and the reason is the one that sank every other rate-based idea in this study: the OpenSky archive carries positions forward between updates, so a *position* is trustworthy whenever it appears, while anything derived by dividing by elapsed time is not. A glideslope is a rate in disguise.

9 What the pipeline actually produces

Every number so far comes from a sweep harness, not from `src/opdi`. The trend sweep re-applies the vote rule to a cached vote table and the endpoint sweeps filter and re-rank a cached candidate table — which is what makes hundreds of parameter cells affordable, and also what makes them unfit, on their own, to justify changing a production default.

So the recommended settings were run through `FlightListProcessor.process_dai` itself and scored the same way. Three differences between harness and pipeline are known in advance and are not bugs:

1. the pipeline selects among candidate aerodromes by **H3 ring count** (`distance_from_center`, an integer stepping about 5.2 km at resolution 7), while the harness uses exact haversine distance;
2. the pipeline filters zones by radius **band**, the harness by exact radius;
3. the harness smooths barometric altitude across the flight-level boundary and the pipeline does not.

The gap those produce is the fidelity of every tuned number on this page.

Run	Role	Mode	Coverage	Accuracy	Correct	Wrong	Score
legacy	ADEP	trend	68.72%	98.82%	64,593	772	63,049
legacy	ADES	trend	68.06%	97.87%	63,362	1,376	60,610
trend	ADEP	trend	78.06%	97.22%	72,180	2,063	68,054
trend	ADES	trend	72.17%	91.17%	62,586	6,059	50,468
endpoint	ADEP	endpoint	80.96%	98.16%	75,587	1,416	72,755
endpoint	ADES	endpoint	69.10%	95.63%	62,856	2,870	57,116
nearest	ADEP	nearest	91.68%	87.78%	76,545	10,656	55,233
nearest	ADES	nearest	88.56%	79.41%	66,892	17,344	32,204
combined	ADEP	endpoint	80.90%	98.23%	75,585	1,365	72,855
combined	ADES	trend	68.26%	96.69%	62,778	2,149	58,480
recommended	ADEP	endpoint	80.90%	98.22%	75,585	1,368	72,849
recommended	ADES	trend	66.29%	98.24%	61,946	1,111	59,724

Harness against pipeline, at the same parameters:

Role	Method	Harness score	Pipeline score	Residual	Relative
ADEP	endpoint	72,755	72,755	+0	+0.00%
ADES	trend	61,038	58,480	−2,558	−4.19%

The two rows could hardly be more different, and the difference is the single most important result on this page.

endpoint reproduces exactly. That is not luck: the endpoint sweep filters the same cached candidate table the pipeline reads, applying the same rule, so harness and pipeline are the same computation and agree to the flight.

trend does not. The harness ranks candidate aerodromes by exact haversine distance. The pipeline first keeps only the candidates at the minimum H3 *ring count* — an integer stepping about 5.2 km at resolution 7 — and applies the haversine tie-break, and the scheduled-service penalty, only within that surviving set. An aerodrome one ring further out is gone before the penalty can promote it. So a penalty tuned against exact distance does something weaker in production, and the harness’s trend optimum is optimistic by the margin shown.

9.1 What running the pipeline caught

The first pipeline path run produced a result that no amount of sweeping could have: the scheduled-service penalty step scored **exactly** what the step before it did, to the flight. An identical score is not a weak effect, it is no effect, and a parameter with no effect is a parameter that is not reaching the code.

It was not. The penalty was added to the ranking in `_compute_flight_table`, but the projection a few lines above it dropped the `apt_scheduled` column, so the ranking looked the column up, failed to find it, and fell back to raw distance. A penalty of ten nautical miles and a penalty of zero produced byte-identical flight lists.

Two things made this visible rather than invisible. The pipeline run put the parameter in front of real data, where a sweep harness — which reads a *different* table that does carry the column — could not. And the fallback printed a note saying the column was missing instead of quietly ranking on distance, because a missing column and a zero penalty look identical in the output. The numbers below are from after the fix.

9.2 The trend tuning re-walked inside the pipeline

Because the harness overstates **trend**, the path decomposition was run again through `process_dai` itself — the same four steps, each one a real pipeline run scored against the same ground truth. This is the version to trust.

Step	Coverage	Accuracy	Correct	Wrong	Score	Gain from this step
production constants	68.06%	97.87%	63,362	1,376	60,610	—
+ scheduled-service penalty	68.06%	97.99%	63,440	1,298	60,844	+234
+ flight-level cap	69.92%	96.29%	64,039	2,468	59,103	−1,741

	Step	Coverage	Accuracy	Correct	Wrong	Score	Gain from this step
	+ vote margin	70.17%	96.25%	64,235	2,506	59,223	+120
	+ radius	69.83%	96.48%	64,084	2,338	59,408	+185

The tuning does not transfer. Inside the pipeline the full recommended arrival setting scores $-1,202$ against the production constants, where the harness predicted $+1,716$ — opposite signs. Every number in Section 4 is measured on a ranking rule the pipeline does not use, and for **trend** that difference is larger than the effect being tuned.

The steps do not fail together. Taken in order, these gain ground: scheduled-service penalty, vote margin, radius; and these lose it: flight-level cap. Only the first step’s figure is a standalone verdict, though — each later step is measured on top of the ones before it, so the vote margin and radius gains are recovering part of what the cap gave away rather than showing that either is worth adopting on its own.

The one that matters is the scheduled-service penalty, worth $+234$ on its own — and the flight-level cap, which the harness liked most of all, is precisely the step that loses most here ($-1,741$). Raising the cap admits state vectors from higher up, where the aircraft is further out, more aerodromes fall in the same H3 ring, and the tie-break the pipeline uses to separate them is coarser than the one the sweep was scored on. So it answers more often ($+1.86$ pp coverage) and is right less often (-1.71 pp accuracy) — a trade the harness’s exact-distance ranking never had to make.

What is adopted for arrivals, therefore, is the production geometry with the scheduled-service penalty added — the only tuned value that is worth anything in the code that will actually run.

10 Why each value, and not its neighbour

A recommendation is only as good as the evidence that picked it, so every value this study chooses appears below with the score at that value, the score at the adjacent grid points, and the score at whatever production uses today. The last column is the one that matters most for trusting the number: a value on a broad **plateau** is a safe recommendation even from a single sample, because its neighbours score nearly the same and a little noise cannot move it far. A value on a sharp isolated **peak** may be partly the sample’s noise, and is flagged as such however good its score.

Role	Parameter	Original	Chosen	Score at original	Score at chosen	Neighbour below	Neighbour above	Δ vs original	Shape
ADES	trend FL cap	40	60	60,720	61,038	60,720	60,259	+318	peak
ADES	trend vote margin	4	2	60,843	61,038	60,996	60,843	+195	plateau
ADES	trend radius (NM)	30	20	61,023	61,038	59,740	61,023	+15	plateau
ADEP	endpoint radius (NM)	40	30	72,541	72,755	72,591	72,541	+214	plateau

Role	Parameter	Original	Chosen	Score at original	Score at chosen	Neighbour below	Neighbour above	Δ vs original	Shape
ADEP	endpoint height (ft)	15000	15000	72,755	72,755	71,591	72,370	+0	peak

The penalty is treated separately because it is not a geometric parameter: it re-ranks candidates without changing how many are answered, so its sweep holds the geometry fixed and appears in Section 6.

10.1 Which change is actually doing the work

The table above varies one parameter at a time around the chosen setting, which shows whether each value is defensible but not how much each contributes to the total gain. This walks the whole distance instead: from the production constants to the recommended setting, changing one parameter per step.

	Step	Score	Gain from this step	Share of total gain
production constants (FL40, margin 4, 30 NM, no penalty)		59,322	—	—
+ scheduled-service penalty of 10 NM		60,216	+894	52%
+ flight-level cap FL60		60,830	+614	36%
+ vote margin 2		61,023	+193	11%
+ radius 20 NM		61,038	+15	1%

The gain is not spread evenly. **scheduled-service penalty of 10 NM** accounts for 52% of it on its own. That is worth stating plainly: the single most valuable change to arrival detection is not a tuned threshold at all, but applying to **trend** the scheduled-service tie-break that **endpoint** already had and **trend** never did.

By contrast the radius change is worth 15 flights out of 61,038 — well inside what one three-day sample can resolve. It is reported as *not motivated*: the sweep’s argmax prefers it, but the evidence does not justify moving a published constant for it.

11 The verdict

Configuration	ADEP	ADES	ADEP coverage	ADEP accuracy	ADEP score	ADES coverage	ADES accuracy	ADES score	Total
recommended	endpoint	trend	80.90%	98.22%	72,849	66.29%	98.24%	59,724	132,573
combined	endpoint	trend	80.90%	98.23%	72,855	68.26%	96.69%	58,480	131,335
endpoint	endpoint	endpoint	80.96%	98.16%	72,755	69.10%	95.63%	57,116	129,871
legacy	trend	trend	68.72%	98.82%	63,049	68.06%	97.87%	60,610	123,659
trend	trend	trend	78.06%	97.22%	68,054	72.17%	91.17%	50,468	118,522
nearest	nearest	nearest	91.68%	87.78%	55,233	88.56%	79.41%	32,204	87,437

The recommended configuration scores **132,573 against production’s 123,659, a gain of +8,914** on a three-day sample of 95,116 flights. Every figure in that table is a real pipeline run scored against Network Manager ground truth.

11.1 What is being recommended, and why

Departures — endpoint, 30 NM, 15,000 ft, 10 NM scheduled-service penalty. Score 72,849 against production’s 63,049, a gain of +9,800. This is the clearest result in the study: the setting is an interior optimum in both parameters, the method beats **trend** at every exchange rate tried, and the pipeline reproduces the sweep exactly, so the tuned figure is the production figure.

Arrivals — trend at the production geometry, with the scheduled-service penalty added. Measured on its own, that is 60,844 against production’s 60,610, a gain of +234. None of the other tuned arrival values is adopted, because none of them survives the pipeline’s ranking rule.

Reported in the combined flight list the arrival half comes out at 59,724, which is −886 against production rather than the +234 just quoted. The difference is not the algorithm. Taking departures from **endpoint** adds tracks that **trend** alone never produced — the merged list holds 233,144 rows against 113,999 — and where two tracks share an aircraft, callsign and date, the scoring alignment awards the flight to whichever starts closest to the actual off-block time. Sometimes that is now an endpoint-only track carrying no arrival. The merge itself loses nothing: both tables name the same 85,744 arrivals, and they disagree on none of them.

That distinction is worth being exact about, because it decides whether the arrival figure is a cost or an artefact. The two tables carry **the same arrival column**: every track id that has an ADES in one has the same ADES in the other, and there are no disagreements in either direction. So the arrival movements OPDI would publish are identical — an analyst counting arrivals at any aerodrome gets the same number from either table. What changes is only which track this benchmark pairs with each reference flight when several are eligible.

So the recommendation gains +9,800 on the departure side and leaves the arrival side materially unchanged: +234 from the penalty, measured like for like, and no change at all to the arrivals actually named. The −886 reported against the combined list is the benchmark’s alignment, not the algorithm’s output, and it is shown rather than suppressed because the same artefact appears in the per-aerodrome table below.

The penalty was checked separately at aerodrome level, where a re-ranking rule could easily rob one aerodrome to pay another. Across the hundred busiest it does not:

Role	Recall without penalty	Recall with penalty	Improved	Worsened	Unchanged
Departures	83.40%	83.60%	4	0	96
Arrivals	79.75%	79.84%	5	0	95

No aerodrome in the hundred is made worse by it. That is the signature of a tie-break that is breaking ties in the right direction rather than trading one error for another.

11.2 The traffic library's rule, for reference

nearest — the unconditional rule the **traffic** library uses — answers far more often than anything else here (91.68% of departures, 88.56% of arrivals) and is right far less often (87.78% and 79.41%). Scored, it is last by a wide margin: 87,437 against the recommendation's 132,573.

That is not a criticism of **traffic**, which is solving a different problem: it names the nearest aerodrome and leaves the judgement to the caller. It is, however, the measurement that justifies OPDI carrying its own rule rather than adopting that one.

12 By aerodrome class

The candidate set is large and medium aerodromes only, so a flight whose true aerodrome is a small airfield or a heliport cannot be named correctly by any method on this page. This is the size of that population, and of everything else.

Departures

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,661	331	71,174	70,597	88.24%	99.19%
out of area	7,892	1	914	626	11.58%	68.49%
medium_airport	6,069	352	4,705	4,362	77.53%	92.71%
small_airport	361	76	132	0	36.57%	0.00%
heliport	133	65	28	0	21.05%	0.00%

Arrivals

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,683	331	59,461	58,591	73.70%	98.54%
out of area	7,964	1	20	0	0.25%	0.00%
medium_airport	5,983	346	3,483	3,355	58.21%	96.33%
small_airport	352	70	83	0	23.58%	0.00%
heliport	134	58	10	0	7.46%	0.00%

Outside the candidate set — 3 classes, 16,836 flights across both roles — the method is working against a reference it was never given. Those flights are counted in the denominator throughout this page, so every coverage figure here already carries their cost.

13 The busiest 100 aerodromes

Movement counts are what OPDI is for, so the last check is per aerodrome rather than per flight: for each of the hundred busiest in the reference, how many movements the recommended methodology finds, against how many the Network Manager data says there were, and how the algorithm production runs today compares on the same aerodromes.

Count ratio is movements found over movements in truth — the number a capacity or delay analysis would be distorted by. **Recall** is the share of that aerodrome's true movements correctly attributed to it.

13.1 Departures

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
1	LTFM	2,282	2,205	2,198	0.967	96.32%	66.70%	+29.62 PP
2	EHAM	2,148	2,143	2,143	1.005	99.77%	99.81%	-0.05 PP
3	EDDF	2,094	2,082	2,082	0.995	99.43%	96.94%	+2.48 PP
4	LFPG	2,084	2,064	2,060	0.993	98.85%	99.04%	-0.19 PP
5	EGLL	2,001	1,989	1,988	1.011	99.35%	99.25%	+0.10 PP
6	LEMD	1,773	1,760	1,759	0.992	99.21%	94.59%	+4.62 PP
7	LEBL	1,591	1,580	1,579	0.992	99.25%	99.43%	-0.19 PP
8	EDDM	1,573	1,564	1,564	0.996	99.43%	99.36%	+0.06 PP
9	LIRF	1,449	1,444	1,444	0.997	99.65%	99.72%	-0.07 PP
10	LTAI	1,435	1,376	1,375	0.963	95.82%	61.25%	+34.56 PP
11	LEPA	1,417	1,400	1,400	0.989	98.80%	92.59%	+6.21 PP
12	LGAV	1,306	1,239	1,239	0.958	94.87%	23.28%	+71.59 PP
13	EGKK	1,223	1,216	1,216	0.996	99.43%	99.59%	-0.16 PP
14	EIDW	1,185	1,164	1,163	0.981	98.14%	98.23%	-0.08 PP
15	LSZH	1,178	1,171	1,164	0.991	98.81%	98.47%	+0.34 PP
16	LOWW	1,156	1,135	1,134	0.988	98.10%	98.88%	-0.78 PP
17	EKCH	1,140	1,132	1,132	0.994	99.30%	98.07%	+1.23 PP
18	LTFJ	1,137	1,102	1,101	0.980	96.83%	93.58%	+3.25 PP
19	LPPT	1,010	1,001	1,001	0.991	99.11%	80.00%	+19.11 PP
20	LFPO	992	982	982	0.997	98.99%	99.29%	-0.30 PP
21	LIMC	973	960	959	0.987	98.56%	99.28%	-0.72 PP
22	ENGM	932	922	922	0.995	98.93%	91.85%	+7.08 PP
23	EGCC	926	903	902	0.974	97.41%	99.35%	-1.94 PP

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
24	EGSS	885	882	882	0.997	99.66%	99.32%	+0.34
25	EPWA	875	871	870	0.999	99.43%	99.20%	+0.23
26	EBBR	872	869	869	1.009	99.66%	99.54%	+0.11
27	EDDB	870	852	850	0.993	97.70%	97.47%	+0.23
28	LEMG	850	847	847	0.999	99.65%	99.53%	+0.12
29	ESSA	820	791	787	0.965	95.98%	96.34%	-0.37
30	EDDL	758	746	744	0.989	98.15%	99.08%	-0.92
31	LFMN	707	699	693	0.982	98.02%	97.74%	+0.28
32	LSGG	680	676	676	1.010	99.41%	67.79%	+31.62
33	LKPR	667	656	655	0.988	98.20%	98.65%	-0.45
34	EFHK	659	653	653	1.002	99.09%	99.24%	-0.15
35	LEAL	599	598	598	0.998	99.83%	99.83%	+0.00
36	LHBP	578	555	553	0.958	95.67%	98.10%	-2.42
37	EGPH	577	570	569	0.986	98.61%	98.96%	-0.35
38	EGGW	550	549	549	1.002	99.82%	98.00%	+1.82
39	EDDH	538	531	528	0.998	98.14%	98.33%	-0.19
40	GCLP	520	23	0	0.000	0.00%	0.00%	+0.00
41	EDDK	515	501	501	0.975	97.28%	96.89%	+0.39
42	LLBG	513	501	501	0.992	97.66%	97.08%	+0.58
43	LPPR	513	510	507	0.988	98.83%	99.03%	-0.19
44	LROP	504	493	493	0.978	97.82%	98.02%	-0.20
45	LEIB	495	490	490	0.990	98.99%	89.49%	+9.49
46	EGBB	474	468	468	0.987	98.73%	98.52%	+0.21
47	LIML	458	451	451	0.987	98.47%	98.47%	+0.00

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
48	LIME	453	450	450	0.993	99.34%	98.68%	+0.66 pp
49	LIRN	449	381	335	0.751	74.61%	0.00%	+74.61 pp
50	LFML	444	440	439	0.989	98.87%	99.32%	-0.45 pp
51	EDDS	440	421	415	0.950	94.32%	93.41%	+0.91 pp
52	LGIR	424	6	0	0.000	0.00%	0.00%	+0.00 pp
53	LIPZ	412	408	408	0.993	99.03%	98.79%	+0.24 pp
54	LFLL	411	402	402	1.015	97.81%	98.78%	-0.97 pp
55	LPFR	405	401	399	0.985	98.52%	98.77%	-0.25 pp
56	LYBE	394	389	389	0.990	98.73%	98.22%	+0.51 pp
57	LICC	390	385	381	0.979	97.69%	0.00%	+97.69 pp
58	LTAC	381	7	0	0.000	0.00%	0.00%	+0.00 pp
59	LIPE	380	375	375	0.987	98.68%	98.42%	+0.26 pp
60	LCLK	368	353	351	0.957	95.38%	86.96%	+8.42 pp
61	GMMN	360	320	315	1.069	87.50%	62.78%	+24.72 pp
62	EPKK	360	349	347	0.967	96.39%	98.89%	-2.50 pp
63	EGGD	359	356	355	0.989	98.89%	99.44%	-0.56 pp
64	LEVC	355	342	341	0.961	96.06%	90.42%	+5.63 pp
65	LFSB	354	351	349	0.994	98.59%	98.31%	+0.28 pp
66	ENBR	349	343	343	1.026	98.28%	96.85%	+1.43 pp
67	LTBJ	346	324	275	0.795	79.48%	1.45%	+78.03 pp
68	EGPF	344	324	322	0.948	93.60%	93.60%	+0.00 pp
69	LGRP	340	327	326	0.976	95.88%	93.82%	+2.06 pp
70	LMML	326	324	324	0.997	99.39%	98.77%	+0.61 pp
71	GCTS	321	5	0	0.019	0.00%	0.00%	+0.00 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
72	GCXO	319	306	306	1.141	95.92%	22.26%	+73.67 pp
73	LATI	316	7	0	0.000	0.00%	0.00%	+0.00 pp
74	LICJ	310	287	287	0.929	92.58%	53.87%	+38.71 pp
75	EBCI	304	304	304	1.000	100.00%	100.00%	+0.00 pp
76	BIKF	297	296	294	1.084	98.99%	98.32%	+0.67 pp
77	ELLX	294	292	292	0.993	99.32%	98.64%	+0.68 pp
78	LFBO	293	289	289	0.993	98.63%	98.98%	-0.34 pp
79	LGTS	279	271	270	0.971	96.77%	68.82%	+27.96 pp
80	GMMX	278	93	86	0.317	30.94%	0.36%	+30.58 pp
81	LEZL	275	274	273	0.993	99.27%	99.64%	-0.36 pp
82	LTBS	273	219	219	0.802	80.22%	0.00%	+80.22 pp
83	EVRA	273	271	270	0.993	98.90%	98.17%	+0.73 pp
84	LBSF	272	165	165	0.607	60.66%	0.00%	+60.66 pp
85	GCRR	269	1	0	0.000	0.00%	0.00%	+0.00 pp
86	EGNX	267	263	263	0.985	98.50%	98.88%	-0.37 pp
87	EPKT	259	251	249	0.961	96.14%	93.44%	+2.70 pp
88	HEGN	257	0	0	0.000	0.00%	0.00%	+0.00 pp
89	UGTB	257	165	159	0.774	61.87%	0.00%	+61.87 pp
90	LFRS	256	253	253	0.992	98.83%	78.52%	+20.31 pp
91	EDDP	256	255	255	1.004	99.61%	98.05%	+1.56 pp
92	EPGD	248	54	39	0.157	15.73%	6.45%	+9.27 pp
93	LGKR	244	209	207	0.869	84.84%	0.00%	+84.84 pp
94	HECA	244	62	42	0.172	17.21%	6.97%	+10.25 pp
95	EGAA	239	239	239	1.004	100.00%	76.15%	+23.85 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
96	ENZV	236	163	160	0.737	67.80%	30.51%	+37.29 pp
97	LEBB	234	183	172	0.744	73.50%	0.00%	+73.50 pp
98	LFPB	230	227	227	1.000	98.70%	95.65%	+3.04 pp
99	LIBD	228	1	0	0.000	0.00%	0.00%	+0.00 pp
100	LIEO	228	201	194	0.851	85.09%	8.33%	+76.75 pp

13.2 Arrivals

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
1	LTFM	2,258	1,787	1,563	0.692	69.22%	71.30%	-2.08 pp
2	EHAM	2,136	1,878	1,871	0.876	87.59%	90.17%	-2.57 pp
3	EDDF	2,091	1,778	1,772	0.847	84.74%	87.61%	-2.87 pp
4	LFPG	2,066	1,712	1,711	0.830	82.82%	85.77%	-2.95 pp
5	EGLL	1,986	1,581	1,579	0.795	79.51%	85.45%	-5.94 pp
6	LEMD	1,786	1,535	1,533	0.859	85.83%	90.31%	-4.48 pp
7	LEBL	1,597	1,499	1,497	0.939	93.74%	95.62%	-1.88 pp
8	EDDM	1,571	1,469	1,464	0.933	93.19%	94.33%	-1.15 pp
9	LTAI	1,437	846	846	0.591	58.87%	62.21%	-3.34 pp
10	LEPA	1,436	1,406	1,404	0.978	97.77%	98.05%	-0.28 pp
11	LIRF	1,433	1,255	1,251	0.874	87.30%	90.16%	-2.86 pp
12	LGAV	1,317	172	165	0.125	12.53%	12.53%	+0.00 pp
13	EGKK	1,220	1,100	1,099	0.901	90.08%	92.54%	-2.46 pp
14	EIDW	1,188	1,084	1,083	0.913	91.16%	93.27%	-2.10 pp
15	LSZH	1,160	1,052	1,045	0.907	90.09%	91.98%	-1.90 pp
16	EKCH	1,152	1,064	1,045	0.907	90.71%	91.93%	-1.22 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
17	LOWW	1,145	1,048	1,043	0.913	91.09%	93.54%	-2.45 PP
18	LTFJ	1,138	984	976	0.860	85.76%	87.52%	-1.76 PP
19	LPPT	1,001	880	880	0.880	87.91%	90.51%	-2.60 PP
20	LFPO	991	940	940	0.950	94.85%	95.76%	-0.91 PP
21	LIMC	967	855	851	0.881	88.00%	91.62%	-3.62 PP
22	ENGM	940	890	889	0.952	94.57%	95.21%	-0.64 PP
23	EGCC	924	821	820	0.889	88.74%	90.91%	-2.16 PP
24	EGSS	888	835	825	0.931	92.91%	94.71%	-1.80 PP
25	EPWA	880	789	751	0.856	85.34%	88.64%	-3.30 PP
26	EBBR	871	767	764	0.884	87.72%	89.90%	-2.18 PP
27	LEMG	861	834	834	0.969	96.86%	98.03%	-1.16 PP
28	ESSA	841	727	722	0.866	85.85%	86.21%	-0.36 PP
29	EDDB	840	789	786	0.940	93.57%	94.52%	-0.95 PP
30	EDDL	751	715	713	0.956	94.94%	97.20%	-2.26 PP
31	LFMN	710	684	680	0.975	95.77%	96.48%	-0.70 PP
32	LSGG	670	637	635	0.951	94.78%	96.72%	-1.94 PP
33	EFHK	666	574	573	0.863	86.04%	86.64%	-0.60 PP
34	LKPR	657	604	601	0.916	91.48%	93.76%	-2.28 PP
35	LEAL	599	588	588	0.982	98.16%	98.83%	-0.67 PP
36	EGPH	584	541	541	0.933	92.64%	94.69%	-2.05 PP
37	LHBP	577	522	515	0.898	89.25%	93.59%	-4.33 PP
38	EGGW	552	528	522	0.957	94.57%	95.83%	-1.27 PP
39	EDDH	532	509	507	0.957	95.30%	97.93%	-2.63 PP
40	GCLP	521	0	0	0.000	0.00%	0.00%	+0.00 PP

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
41	LLBG	516	429	423	0.822	81.98%	88.37%	-6.40
42	LPPR	512	479	477	0.932	93.16%	93.75%	0.59
43	LEIB	509	497	497	0.978	97.64%	97.64%	0.00
44	EDDK	509	473	469	0.921	92.14%	95.09%	2.95
45	LROP	509	486	485	0.955	95.28%	97.25%	1.96
46	EGBB	466	418	417	0.895	89.48%	92.70%	3.22
47	LIML	465	457	453	0.978	97.42%	97.63%	0.22
48	LIME	460	448	444	0.965	96.52%	97.83%	1.30
49	LIRN	447	1	0	0.000	0.00%	0.00%	0.00
50	LFML	445	439	438	0.987	98.43%	99.10%	0.67
51	EDDS	433	409	409	0.958	94.46%	95.61%	1.15
52	LGIR	430	0	0	0.000	0.00%	0.00%	0.00
53	LFLI	412	398	398	0.966	96.60%	97.33%	0.73
54	LIPZ	409	388	386	0.949	94.38%	95.35%	0.98
55	LPFR	407	403	402	0.993	98.77%	98.77%	0.00
56	LYBE	395	364	364	0.922	92.15%	94.68%	2.53
57	LICC	391	25	23	0.059	5.88%	5.88%	0.00
58	LIPE	376	370	369	0.981	98.14%	98.94%	0.80
59	LTAC	374	0	0	0.000	0.00%	0.00%	0.00
60	LCLK	367	345	344	0.946	93.73%	97.00%	3.27
61	EGGD	360	334	334	0.931	92.78%	95.28%	2.50
62	LEVC	357	347	346	0.969	96.92%	98.04%	1.12
63	GMMN	357	260	260	0.737	72.83%	78.15%	5.32
64	ENBR	357	340	329	0.950	92.16%	92.16%	0.00

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
65	EPKK	356	343	331	0.933	92.98%	94.66%	-1.69 pp
66	LFSB	352	341	341	0.972	96.88%	97.73%	-0.85 pp
67	LGRP	346	315	314	0.910	90.75%	91.04%	-0.29 pp
68	EGPF	346	300	299	0.873	86.42%	88.44%	-2.02 pp
69	LTBJ	341	0	0	0.000	0.00%	0.00%	+0.00 pp
70	LMML	326	318	317	0.975	97.24%	97.85%	-0.61 pp
71	LATI	323	0	0	0.000	0.00%	0.00%	+0.00 pp
72	GCTS	323	0	0	0.000	0.00%	0.00%	+0.00 pp
73	GCXO	320	0	0	0.000	0.00%	0.00%	+0.00 pp
74	LICJ	314	121	121	0.385	38.54%	38.54%	+0.00 pp
75	EBCI	303	298	295	0.974	97.36%	98.68%	-1.32 pp
76	BIKF	296	90	88	0.297	29.73%	33.11%	-3.38 pp
77	ELLX	294	271	271	0.925	92.18%	96.60%	-4.42 pp
78	LFBO	290	287	285	0.983	98.28%	98.97%	-0.69 pp
79	LGTS	281	257	257	0.915	91.46%	91.81%	-0.36 pp
80	LTBS	280	1	1	0.004	0.36%	0.36%	+0.00 pp
81	GMMX	279	1	0	0.000	0.00%	0.00%	+0.00 pp
82	EVRA	278	228	226	0.820	81.29%	82.37%	-1.08 pp
83	LEZL	276	264	263	0.957	95.29%	99.28%	-3.99 pp
84	GCRR	269	0	0	0.000	0.00%	0.00%	+0.00 pp
85	LBSF	267	3	0	0.000	0.00%	0.00%	+0.00 pp
86	EDDP	261	236	235	0.904	90.04%	93.49%	-3.45 pp
87	HEGN	258	0	0	0.000	0.00%	0.00%	+0.00 pp
88	LFRS	257	251	249	0.969	96.89%	97.28%	-0.39 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, production	Δ recall
89	UGTB	257	47	46	0.183	17.90%	20.23%	-2.33 pp
90	EPKT	257	240	239	0.930	93.00%	93.39%	-0.39 pp
91	EGNX	255	231	229	0.898	89.80%	92.94%	-3.14 pp
92	LIEO	252	55	54	0.214	21.43%	21.43%	+0.00 pp
93	LGKR	247	1	0	0.000	0.00%	0.00%	+0.00 pp
94	EGAA	246	233	217	0.882	88.21%	89.43%	-1.22 pp
95	EPGD	244	9	1	0.012	0.41%	0.41%	+0.00 pp
96	ENZV	236	65	64	0.284	27.12%	27.54%	-0.42 pp
97	LEBB	235	0	0	0.000	0.00%	0.00%	+0.00 pp
98	LTFE	232	180	175	0.759	75.43%	78.02%	-2.59 pp
99	HECA	231	63	20	0.087	8.66%	2.16%	+6.49 pp
100	LIBD	229	1	0	0.000	0.00%	0.00%	+0.00 pp

13.3 Across the hundred

Role	Movements in truth	Recall, recommended	Recall, production	Aerodromes improved	Median count ratio
Departures	65,382	92.37%	83.40%	61 of 100	0.988
Arrivals	65,360	77.86%	79.75%	1 of 100	0.899

A count ratio below one means the aerodrome is under-counted — the usual case, since every abstention is a movement not attributed. Above one means movements are being attributed to an aerodrome that did not see them, which is the more damaging error and the one the scoring penalises at rate k .

The departure row is the result: recall rises by nearly nine points and the median aerodrome is within about one percent of its true movement count.

The arrival row must be read with Section 11 in hand. It falls, and on almost every aerodrome, which looks like a broad regression and is not one. The arrival column of the recommended flight list is identical to production's, track for track — the same aerodromes named for the same flights. What changes is that the merged list also contains departure-only tracks, and this table pairs reference flights with tracks by start time, so some arrivals are now counted against a track that carries none. The measurement moved; the data did not. It is left in the table rather than corrected away, because a benchmark that quietly adjusts its own alignment to flatter a result is worth less than one that shows where it is fragile.

14 What ships

The pipeline's `DetectionConfig` defaults are now the values this study supports. Two were changed, three were deliberately left where production had them, and the difference between those two groups is the whole argument of the page: a value is adopted when a **pipeline** run confirms it, not when a sweep prefers it.

Parameter	Production	Now ships	Verdict
<code>trend_sched_penalty_nm</code>	10	10	Changed. The only tuned trend value the pipeline confirms: +234 score, improves 9 of the hundred busiest aerodromes, worsens none.
<code>endpoint_radius_nm</code>	40	30	Changed. Interior optimum, and the pipeline reproduces the sweep exactly.
<code>endpoint_height_ft</code>	15,000	15,000	Confirmed. The sweep's argmax lands on the value already in use.
<code>trend_max_fl</code>	40	40	Rejected. The sweep preferred FL120 by a wide margin; through the pipeline that setting lost 1,741.
<code>trend_radius_nm</code>	30	30	Rejected. The argmax at 20 NM was worth 15 flights out of 61,038 on one sample.
<code>trend_vote_margin</code>	4	4	Rejected. Its gain was measured on top of a flight-level cap that is not being adopted.

Every OPDI flight list published before this change was built with the production column. `DetectionConfig.legacy()` still returns exactly those values, so anything already released stays reproducible by name — the same principle as the frozen **version** strings on published events. A regression test pins the preset, and a second pins the two rejected values, so restoring an optimum on sweep evidence alone has to argue with a test rather than with a comment.

15 Limitations

15.1 Parameters that were not swept

Five parameters were held fixed by decision rather than by evidence. Naming them matters: a reader who sees seven tuned values and no caveat will reasonably assume the whole configuration was optimised, and it was not.

Parameter	Where	Held at	Why not swept
Smoothing half-window	<code>flights.py</code>	2 samples	Parameterised in config so it is visible and changeable, but a smoothing width interacts with the sample rate rather than the geometry, and the decimation study settled the sample rate separately.
Classification dead band	<code>flights.py</code>	none exists	There is no dead band to sweep. The vote margin already plays that role.
Out-of-area border margin	<code>flights.py</code>	30 NM	Fixes what counts as “outside the observed area”. Changing it redefines the ground truth rather than the method, so a sweep would move the target as well as the aim.
Aerodrome type set	<code>h3_airport_zones.py</code>	large + medium	The candidate set. Widening it is a data-volume decision about the zone table, not a detection threshold.
H3 resolution	<code>config.py</code>	7 (± 0.76 NM)	Changing it invalidates every cached zone table and the <code>distance_from_center</code> ring counts that production selects on.

Two of these carry a cost that this study measures rather than waves at, since “not swept” is not the same as “no consequence”:

- The **type set** means a flight whose true aerodrome is a small airfield or a heliport cannot be named correctly by any method here — it is not in the candidate set at all. The per-class table gives the size of that population.
- The **border margin** bounds how well any method can identify flights that begin or end outside the observed area. The out-of-area recall figures give that directly.

15.2 One sample, and what that does and does not license

The parameters are chosen as the argmax over a grid evaluated on a single three-day period. An argmax over hundreds of cells on one sample will always overstate the gain slightly: some of the winning cell's margin is the sample's noise, and it will not reproduce.

That is the reason for the second period, and for reporting the shape of each optimum rather than only its location. A parameter sitting on a broad plateau is a safe recommendation even from one sample; one sitting on a sharp isolated peak is not, however good its score.